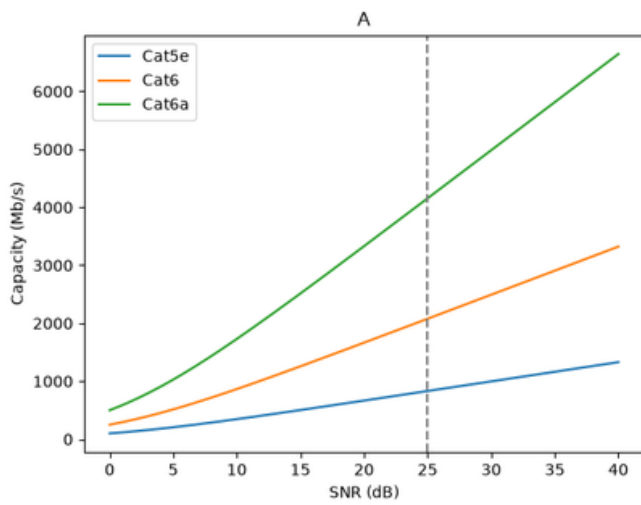
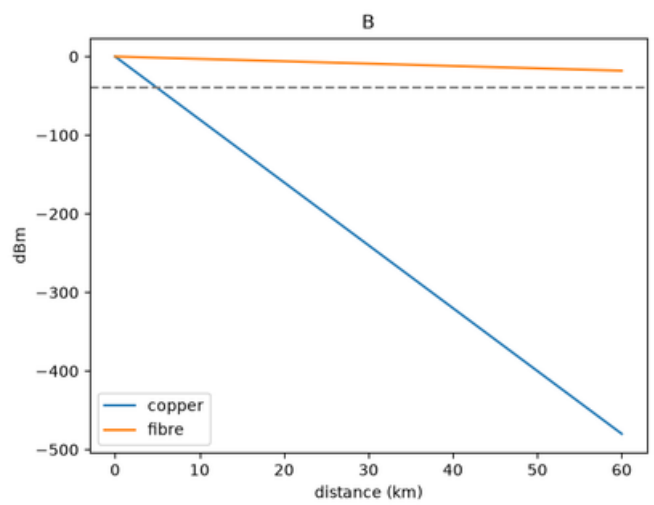


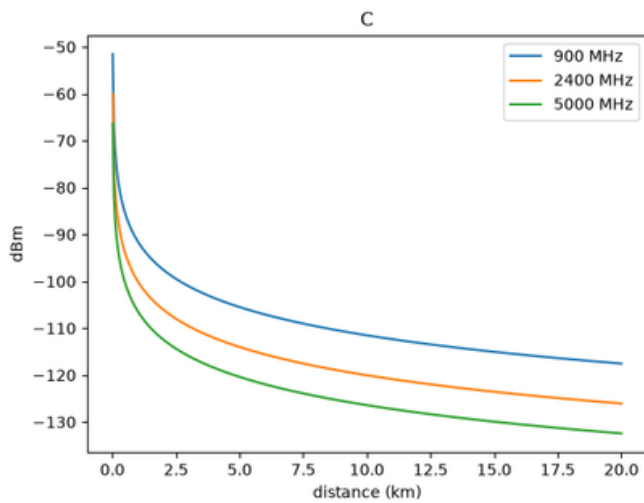
PLOTS



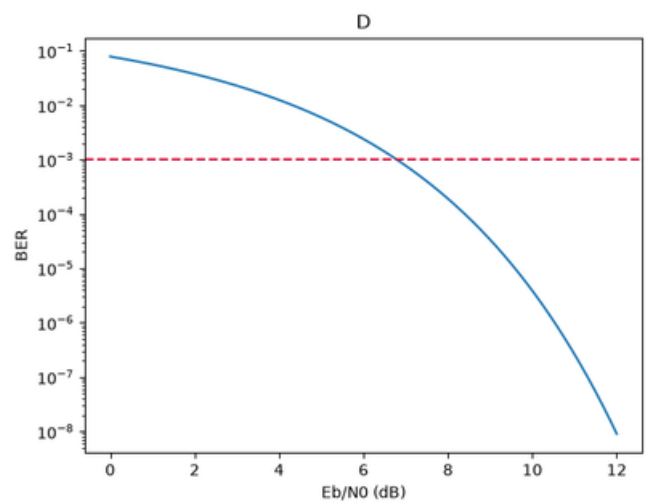
Capacity Vs SNR



Power Vs distance



Wireless Plot



BER Vs Eb/NO

OBSERVATIONS

- Levels for 1 Gb/s per cable:

Cat5e requires 32.0 levels

Cat6 requires 4.0 levels

Cat6a requires 2.0 levels.

- Shannon Capacities:

Cat5e achieves approximately 830.94 Mb/s (830937524.12 B/s)

Cat6 achieves approximately 2077.34 Mb/s (2077343810.30 B/s)

Cat6a achieves approximately 4154.68 Mb/s (4154687620.60 B/s)

- Copper/fibre spans:

The maximum copper span is 5.0 km

The maximum optical fibre span is 133.33 km

- The 100 m derivation:

Derived using a Cat5e attenuation of 220 dB/km at 100 MHz and a link budget of 22 dB, yielding $d_{\max} = 22 \text{ dB} / 220 \text{ dB/km} = 0.1 \text{ km}$, which equates to 100 m.

- One-line channel summary:

The channel features a maximum span of 5 km, a Shannon capacity of 830.94 Mb/s at 25 dB SNR, an operating point of $E_b/N_0=7 \text{ dB}$, and a corresponding Bit Error Rate (BER) of approximately 8×10^{-5} .